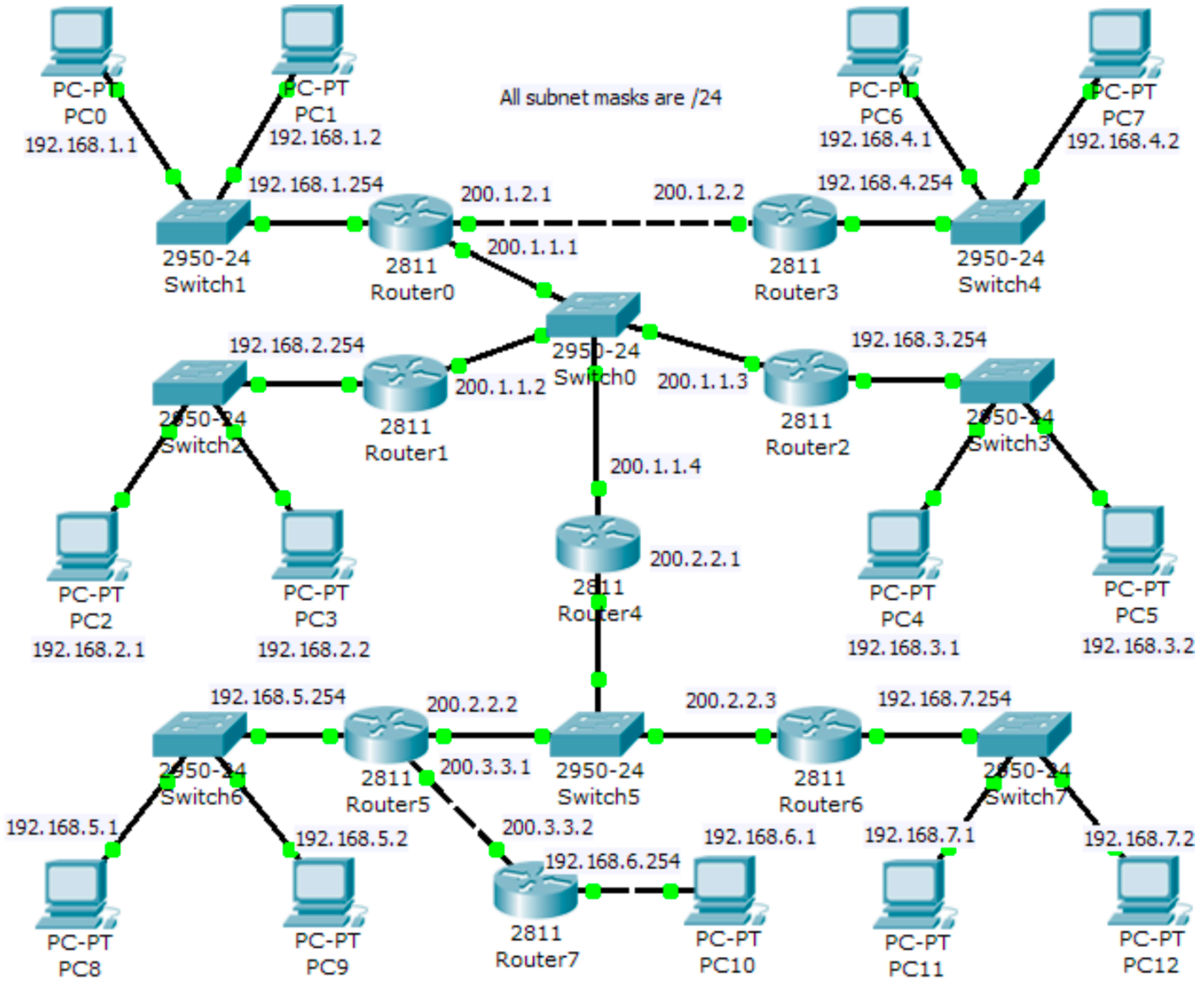
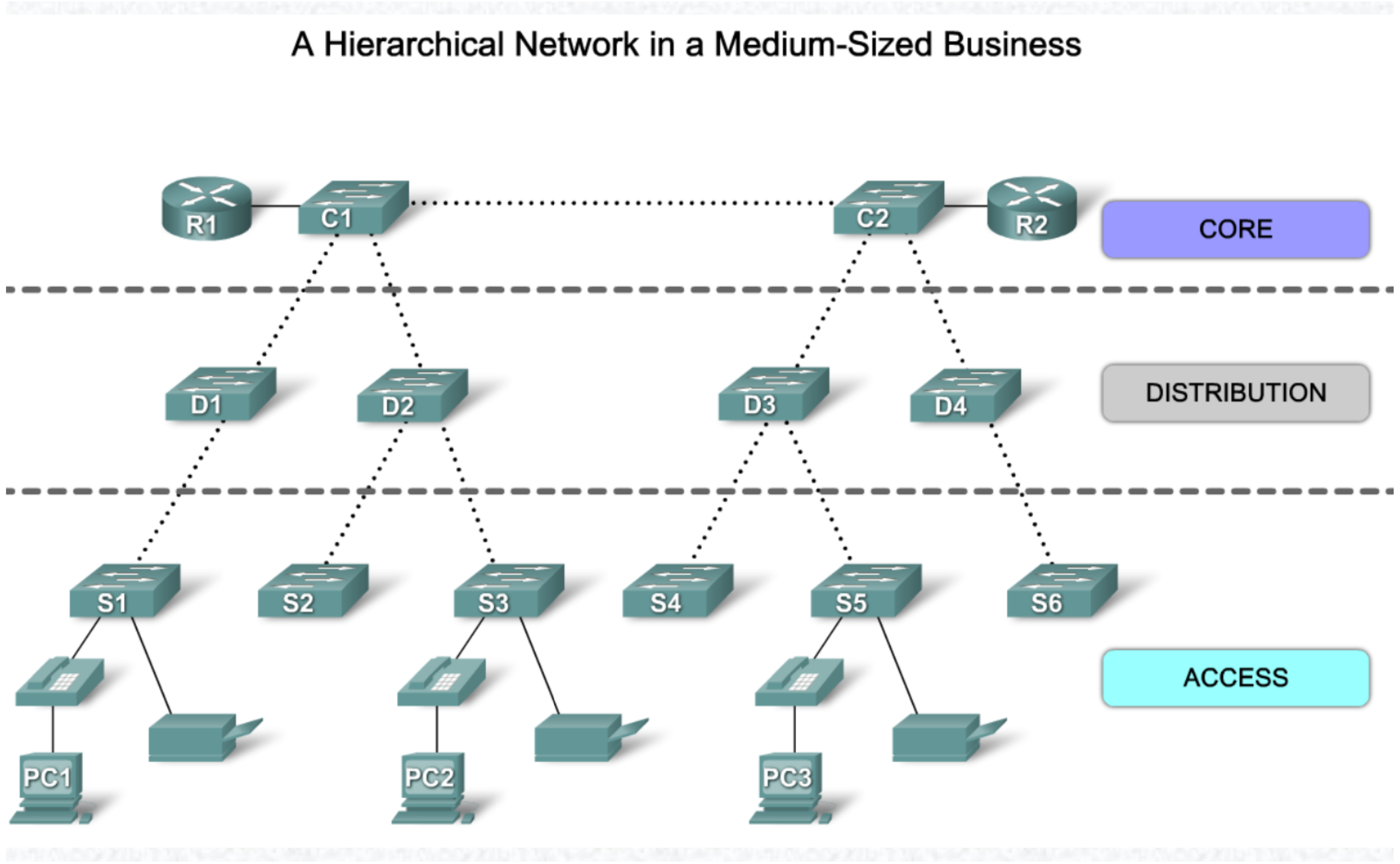
Chapter 6
VIRTUAL LAN





ENTERPRISE NETWORK / CAMPUS NETWORK

WHAT IS VLAN?

- We need **Routers** to build LAN
- We can also use a (layer3) switch to build LAN
- This save costs and also improve network performance
- The kind of network built using switches are call **VLANs**

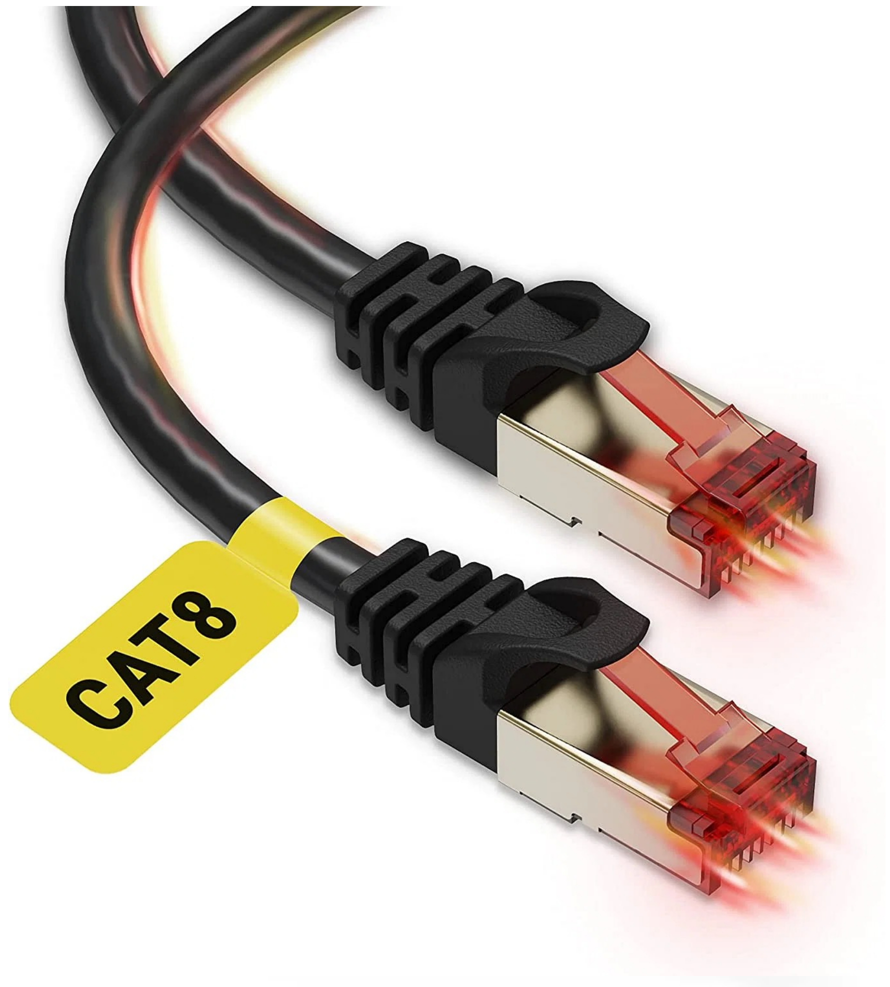
WHY VLAN?

- Routers are more expensive than switches. If we build VLANs with switches, we can save costs and impress our boss.

Build 5 networks using these devices:

VLAN STARTER PACK

Layer 3-7
Network Switch



Cat 6/7/8
Ethernet Cable

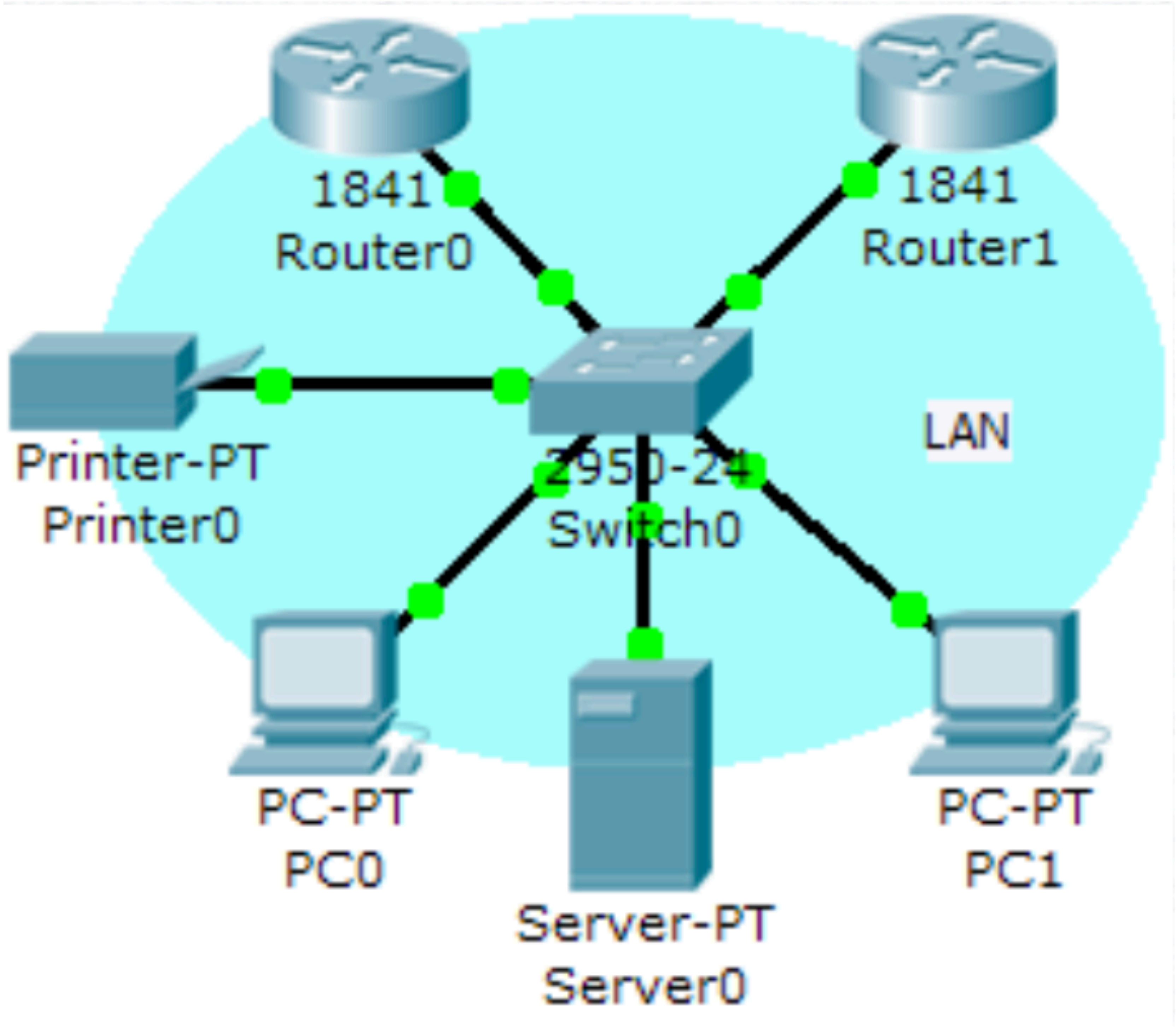
VLAN Config Skills
Spark Joys

Traditional switch (Layer2) only understands MAC address. Layer 3-7 Switch is a more capable switch that understands IP layers and above. We need to use an L3-7 switch to build a

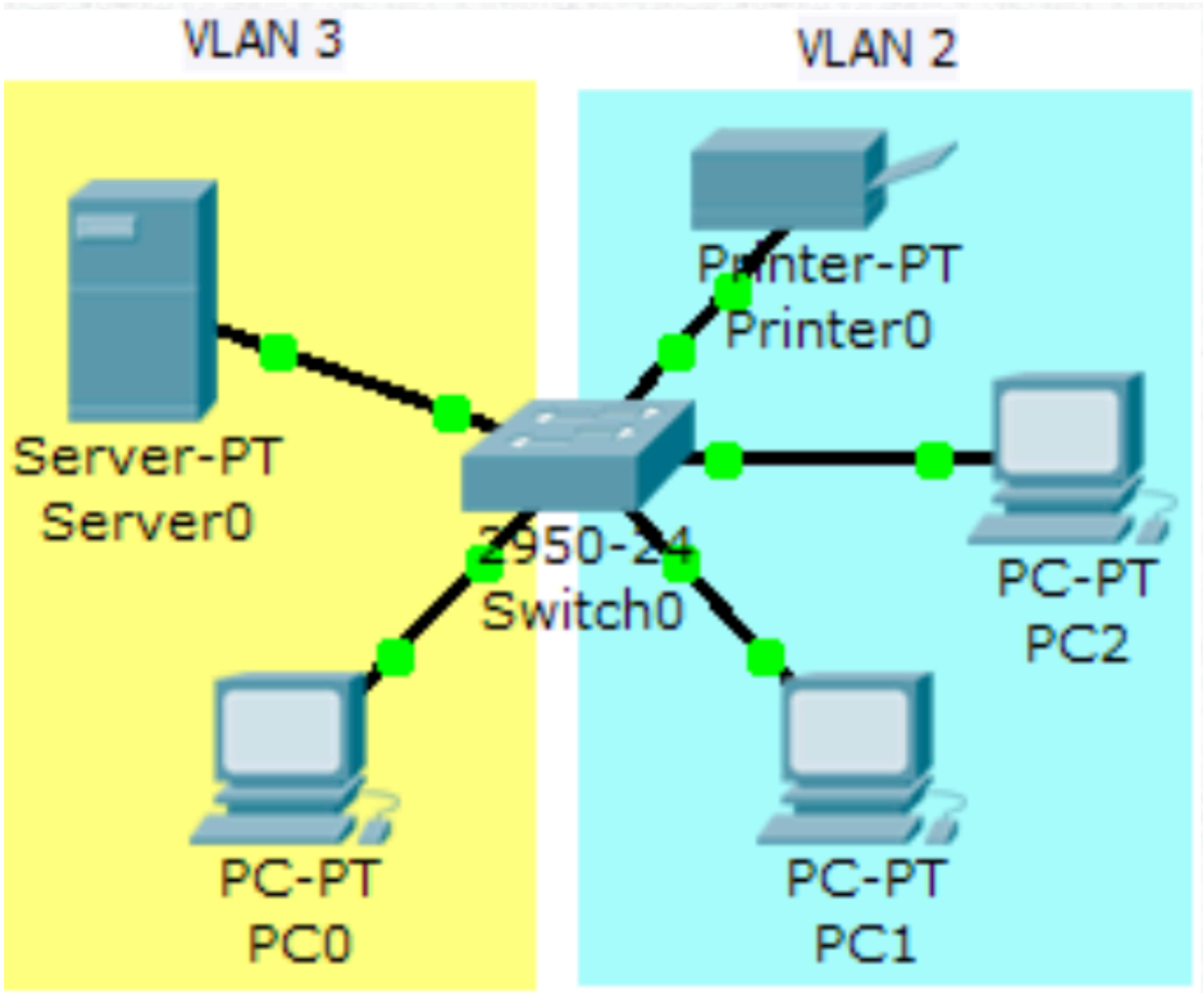
Cat 6/7/8 provides 1Gbps-10Gbps which is suitable for trunking. In virtual LAN, one single unified cable connects multiple VLANs instead of a few cables. This allows us to save the ports on switches. Because of this,

VLAN vs LAN

- LAN – built with routers
- VLAN – built with L3 switches



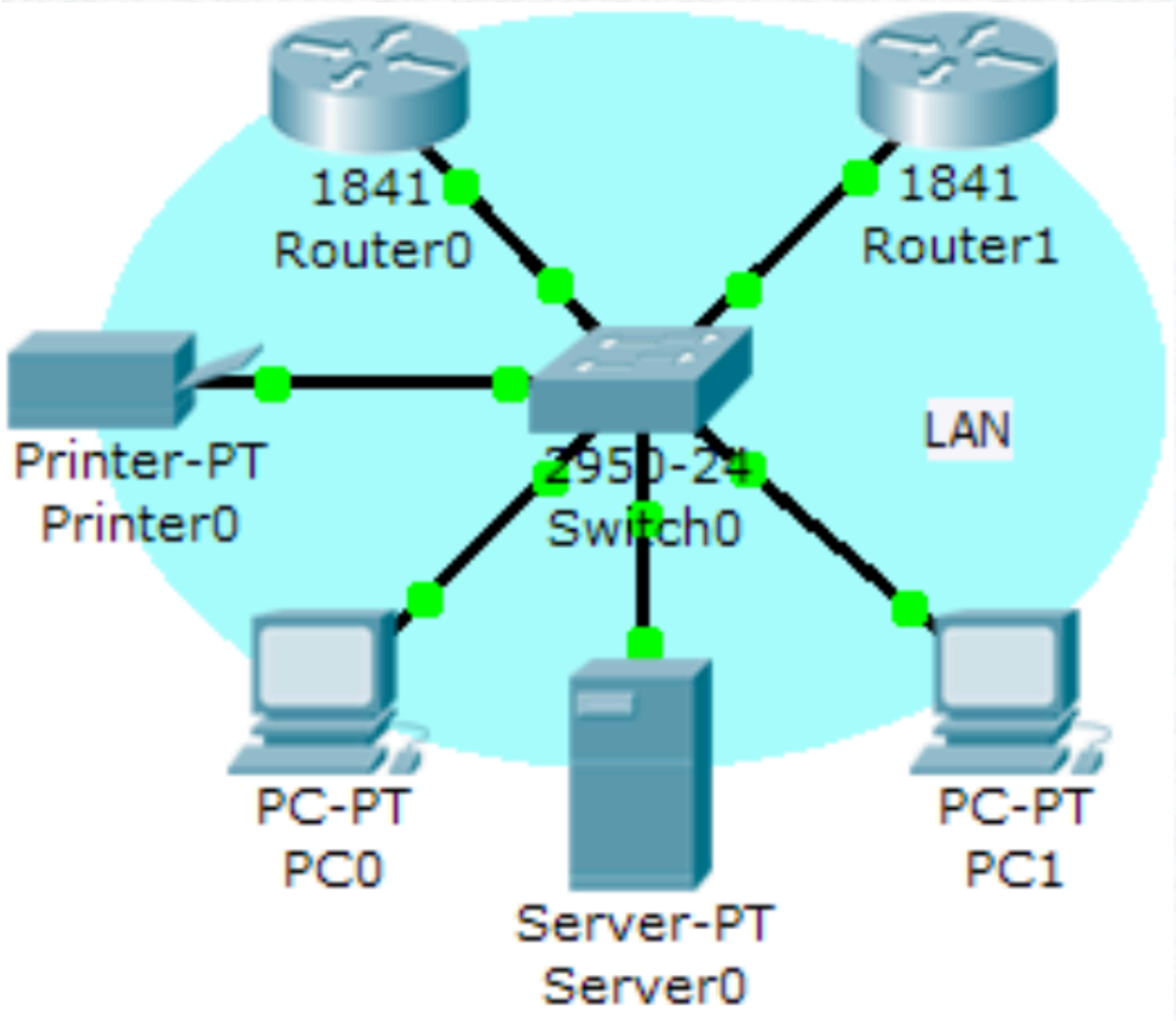
LAN



vLAN

VLAN vs LAN

- Divide these devices into two networks:
  - LAN1: {Server0, PC0}
  - LAN2: {Printer, PC1, PC2}

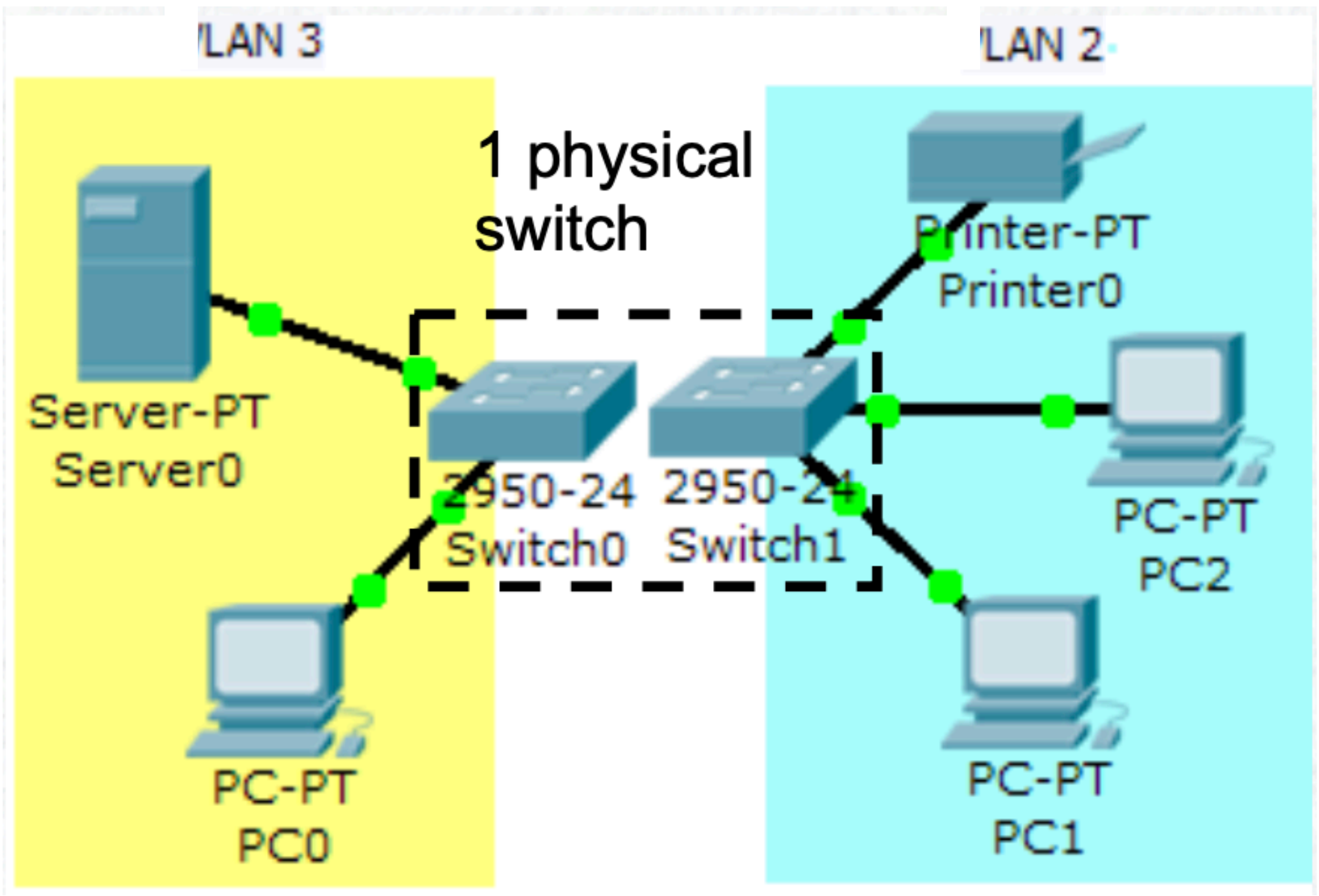


LAN

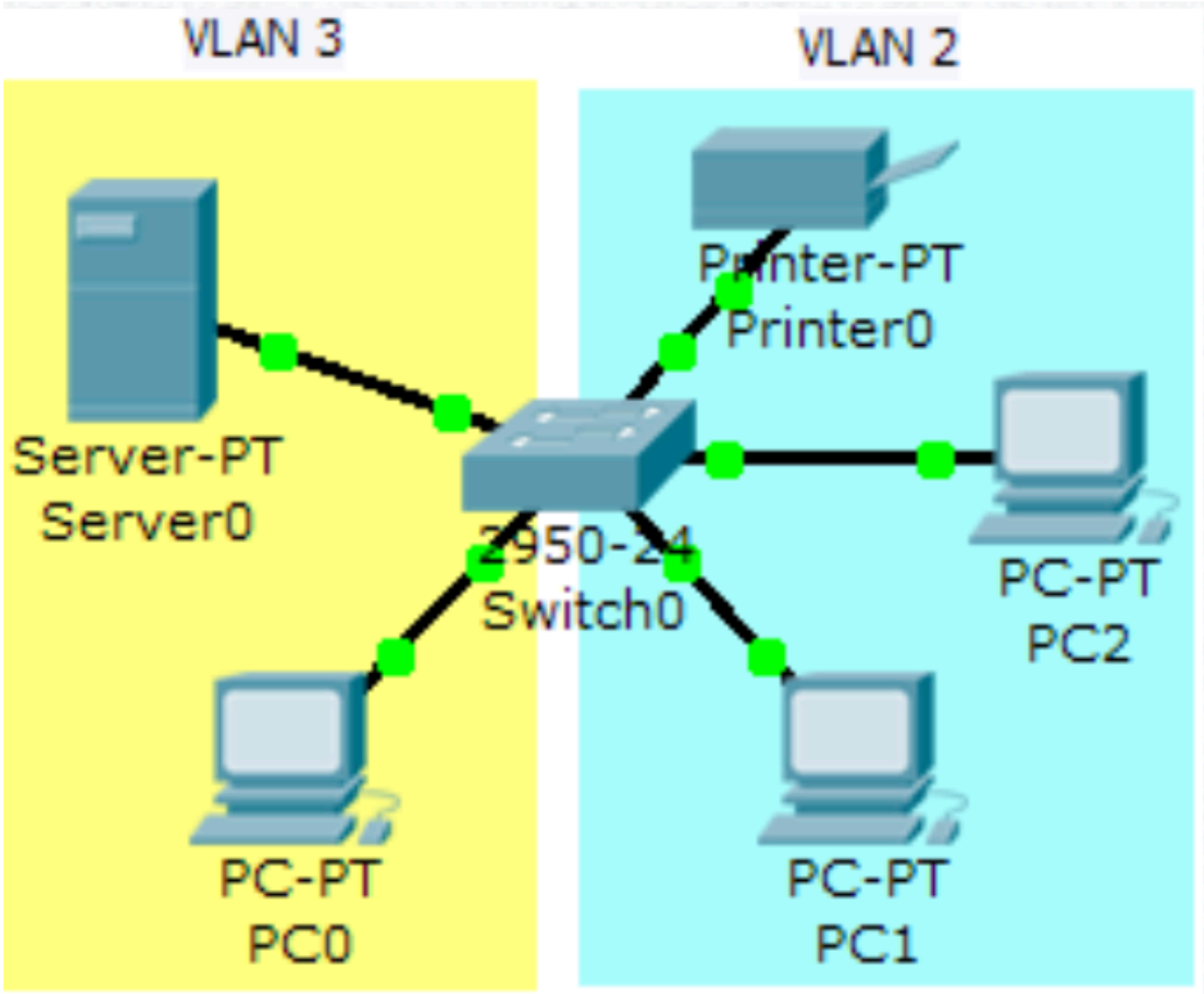
Physical LAN

VLAN vs LAN

- Instead of using two switches, we can make 2 VLANs on the same switch.
- We can go to each port of the switch and configure them into different VLANs.
- For example, if we set port#2 to VLAN2, then, any device connect to port#2 will join VLAN2.



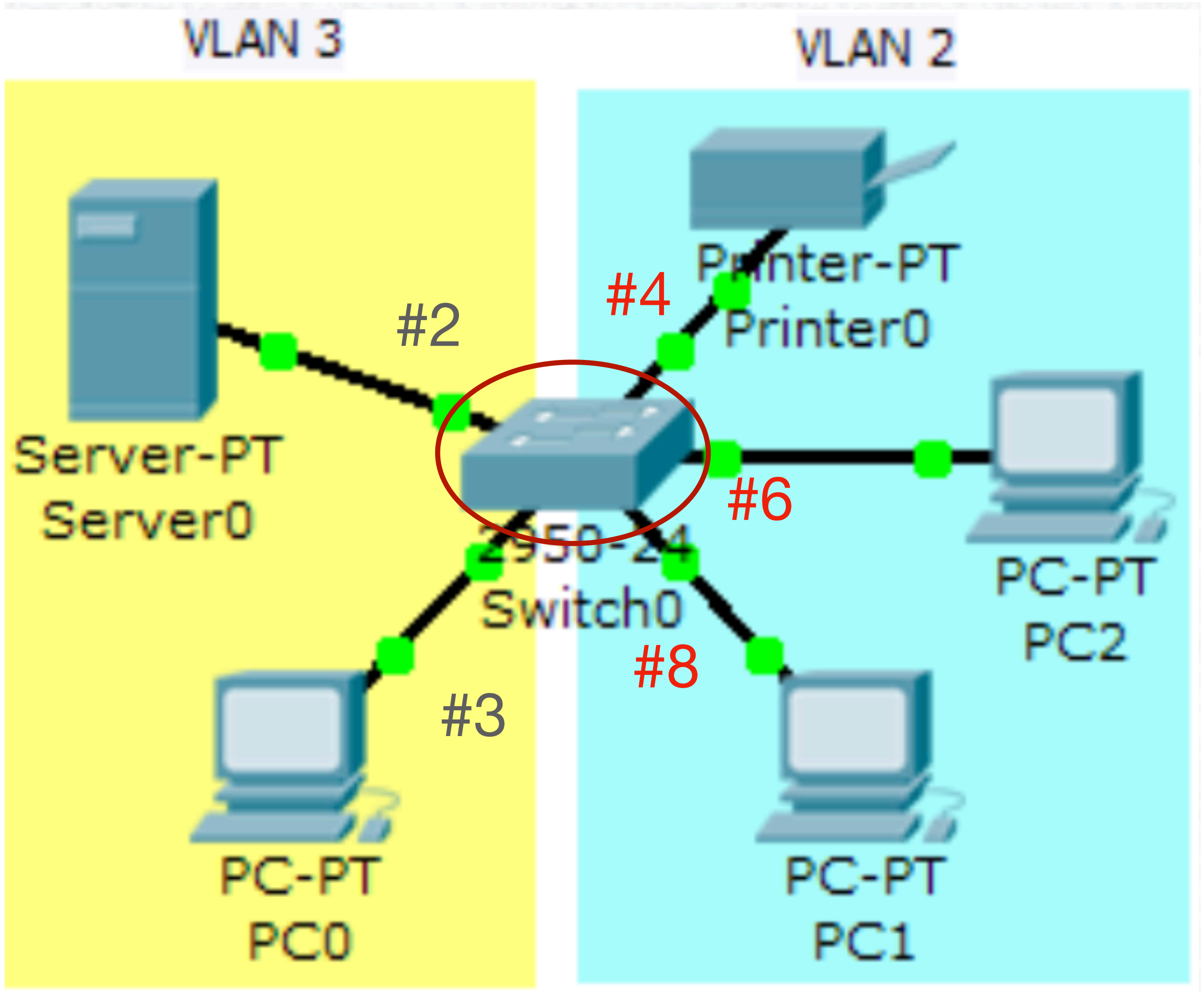
Physical LAN



Virtual LAN

How to set VLAN

- In order to set VLAN, we need to learn how to configure VLAN commands on L3 Switch.
- All the ports on a switch are in VLAN1 by default
- If we did not set anything, all devices will be in VLAN1
- Now we will learn how to set the ports to different VLAN to build more networks.



Virtual LAN

Port#1 V1	Port#2 V3	Port#3 V3	Port#4 V2
Port#5 V2	Port#6 V2	Port#7 V4	Port#8 V2

Switch0

How to set VLAN
(DEMO)

Step1: Create VLAN

```
config t
vlan 2
end

Show vlan br
```

VLAN1 - Default VLAN

VLAN 1002-1005: Token Ring

VLAN 2 - 1001 can be used for your VLAN

```
Switch#conf t
Switch(config)#vlan 2
Switch(config-vlan)#end
Switch#show vlan brief

VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------------------
1    default                          active    Fa0/1, Fa0/2, Fa0/3, Fa0/4
                                                Fa0/5, Fa0/6, Fa0/7, Fa0/8
                                                Fa0/9, Fa0/10, Fa0/11, Fa0/12
                                                Fa0/13, Fa0/14, Fa0/15, Fa0/16
                                                Fa0/17, Fa0/18, Fa0/19, Fa0/20
                                                Fa0/21, Fa0/22, Fa0/23, Fa0/24
2    VLAN0002                         active
1002 fddi-default                     active
1003 token-ring-default               active
1004 fddinet-default                  active
1005 trnet-default                    active
Switch#
```

Create new VLAN.

Verify that the VLAN is created.

Step2: Name VLAN

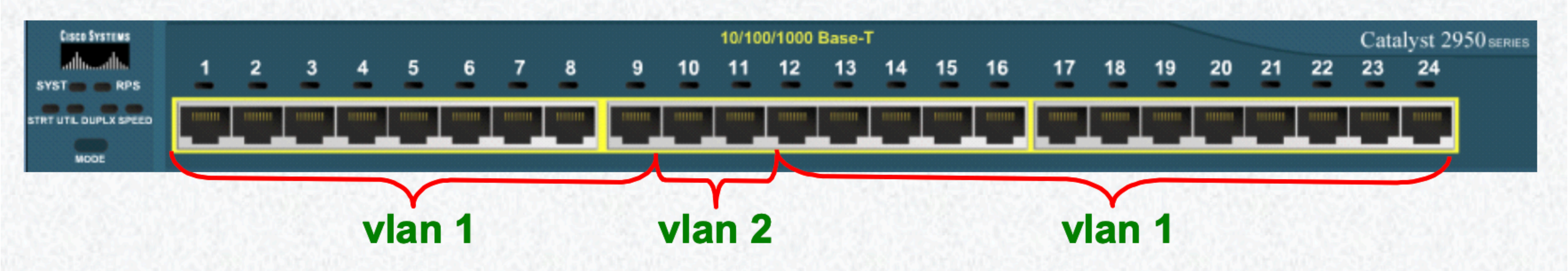
```
config t
vlan 3
name Finance

show vlan br
```

```
Switch(config)#vlan 3
Switch(config-vlan)#name Finance
```

```
VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------------------
1    default                          active    Fa0/1, Fa0/2, Fa0/3, Fa0/5
                                                Fa0/7, Fa0/9, Fa0/10, Fa0/12
                                                Fa0/13, Fa0/14, Fa0/15, Fa0/16
                                                Fa0/17, Fa0/18, Fa0/19, Fa0/20
                                                Fa0/21, Fa0/22, Fa0/23, Fa0/24
3    Finance                          active    Fa0/11
```

Step3: Assign VLAN memberships to PORTs



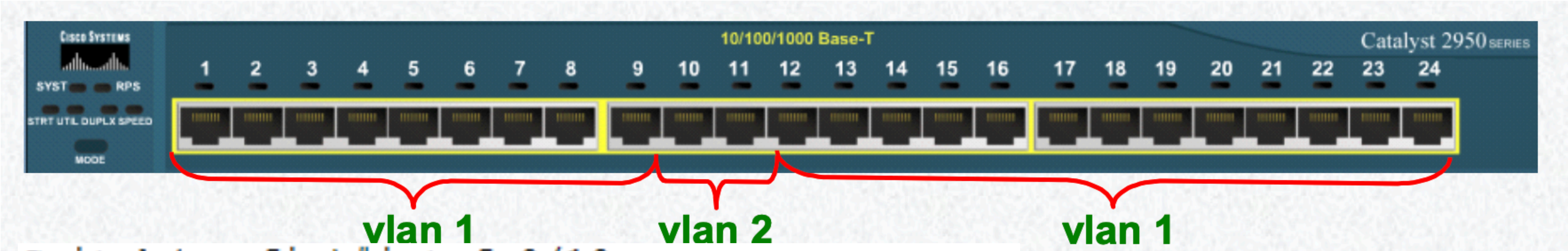

```
Switch(config)#int fa0/10
Switch(config-if)#switchport access vlan 2
Switch(config-if)#exit
Switch(config)#int fa0/11
Switch(config-if)#switchport access vlan 2
Switch(config-if)#exit
```

int range fa0/10-11

```
Switch#show vlan brief

VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------------------
1    default                          active    Fa0/1, Fa0/2, Fa0/3, Fa0/4
                                                Fa0/5, Fa0/6, Fa0/7, Fa0/8
                                                Fa0/9, Fa0/12, Fa0/13, Fa0/14
                                                Fa0/15, Fa0/16, Fa0/17, Fa0/18
                                                Fa0/19, Fa0/20, Fa0/21, Fa0/22
                                                Fa0/23, Fa0/24
2    VLAN0002                         active    Fa0/10, Fa0/11
```

Step4: Delete VLAN (Optional)




```
Switch(config)#int fa0/10
Switch(config-if)#switchport access vlan 2
Switch(config-if)#exit
Switch(config)#int fa0/11
Switch(config-if)#switchport access vlan 2
Switch(config-if)#exit
```

int fa0/11
no switch port access vlan 2

Remove fa0/11 from VLAN2
Fa0/11 will go back to VLAN1

no vlan2

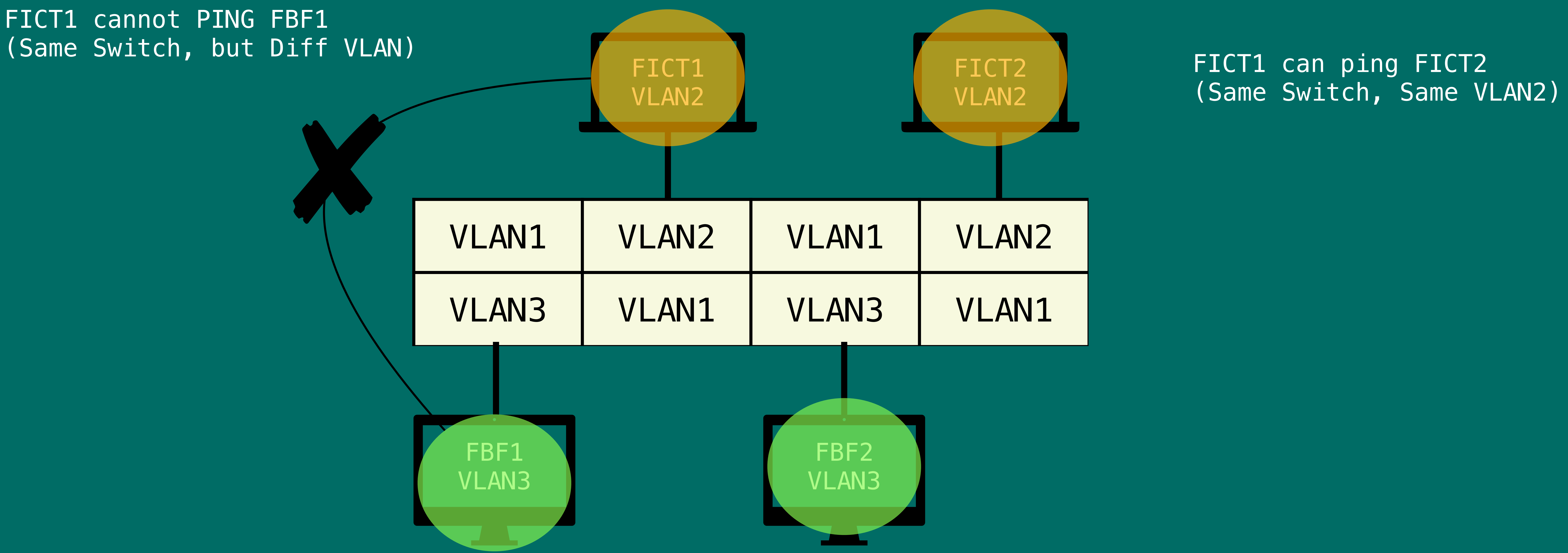
Remove VLAN2 entirely
VLAN2 no more exists
But the PC inside VLAN2 still not automatically go back to VLAN1

VLAN CONNECTIVITY

- VLAN follow the same IP rules (Chapter4) as normal Physical LAN
- Let’s take a look at which PC can/cannot ping based on these setup.

VLANs Connectivity

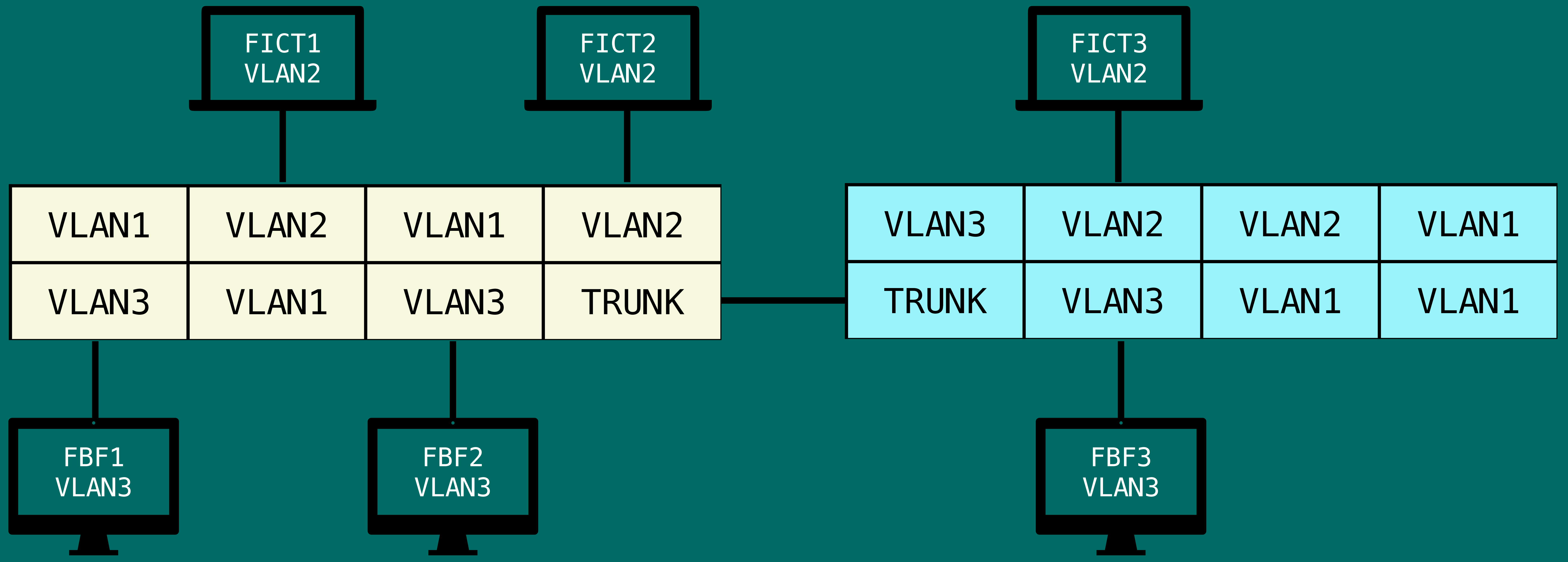
BUILD TWO NETWORKS TO ISOLATE FICT AND FBF



FBF1 can ping FBF2
(Same Switch, Same VLAN3)

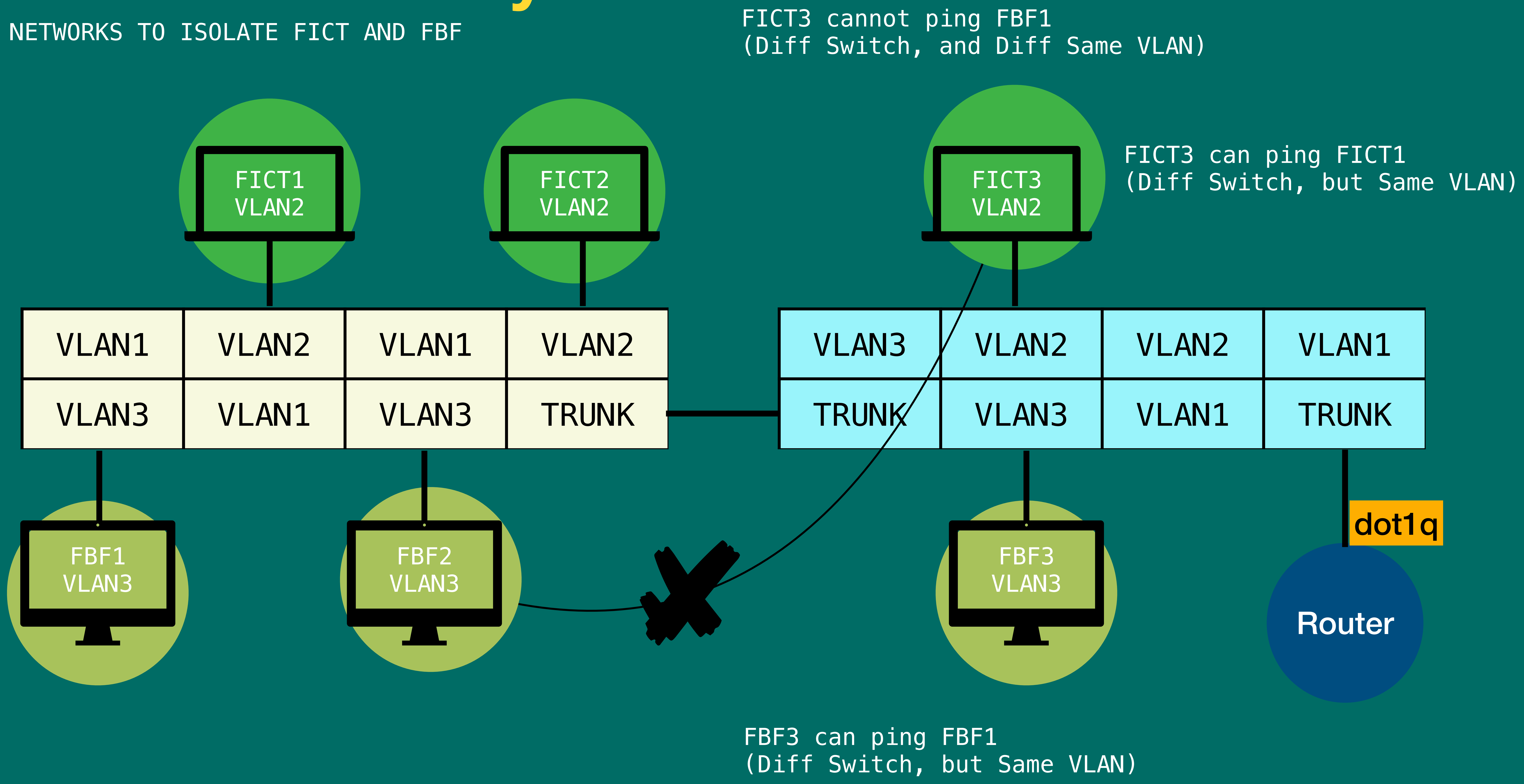
VLANs Connectivity

BUILD TWO NETWORKS TO ISOLATE FICT AND FBF



VLANs Connectivity

BUILD TWO NETWORKS TO ISOLATE FICT AND FBF



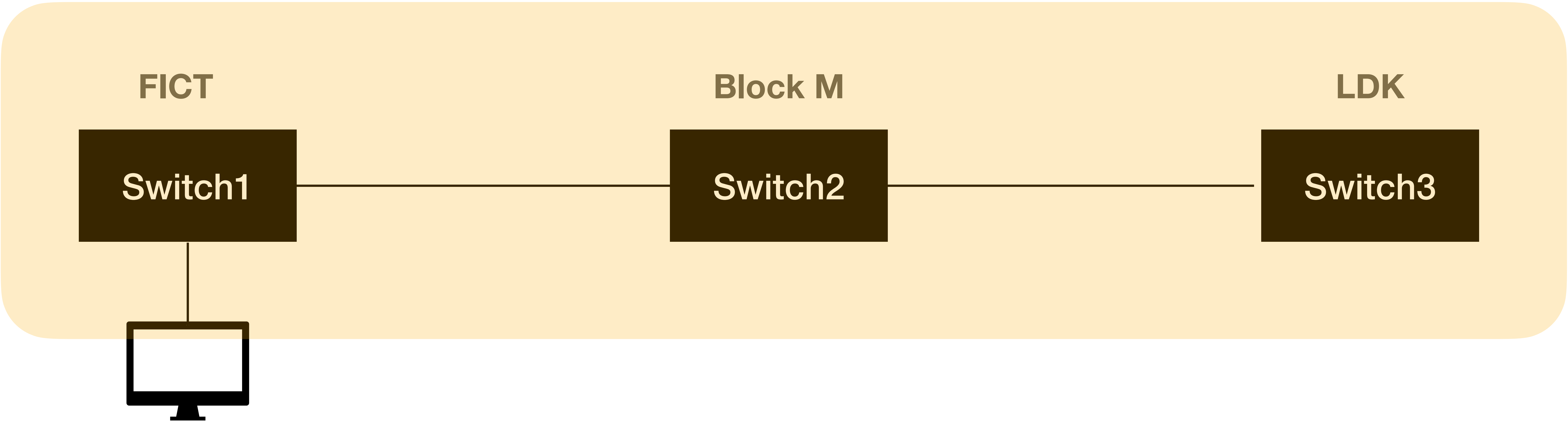
VLAN SPANNING

VLAN SPANNING

- We can build a very large network(s) with VLAN.

- For example, through VLAN Spanning, we can link the Kampar campus to Sg long campus into one network (although they are physically far away).

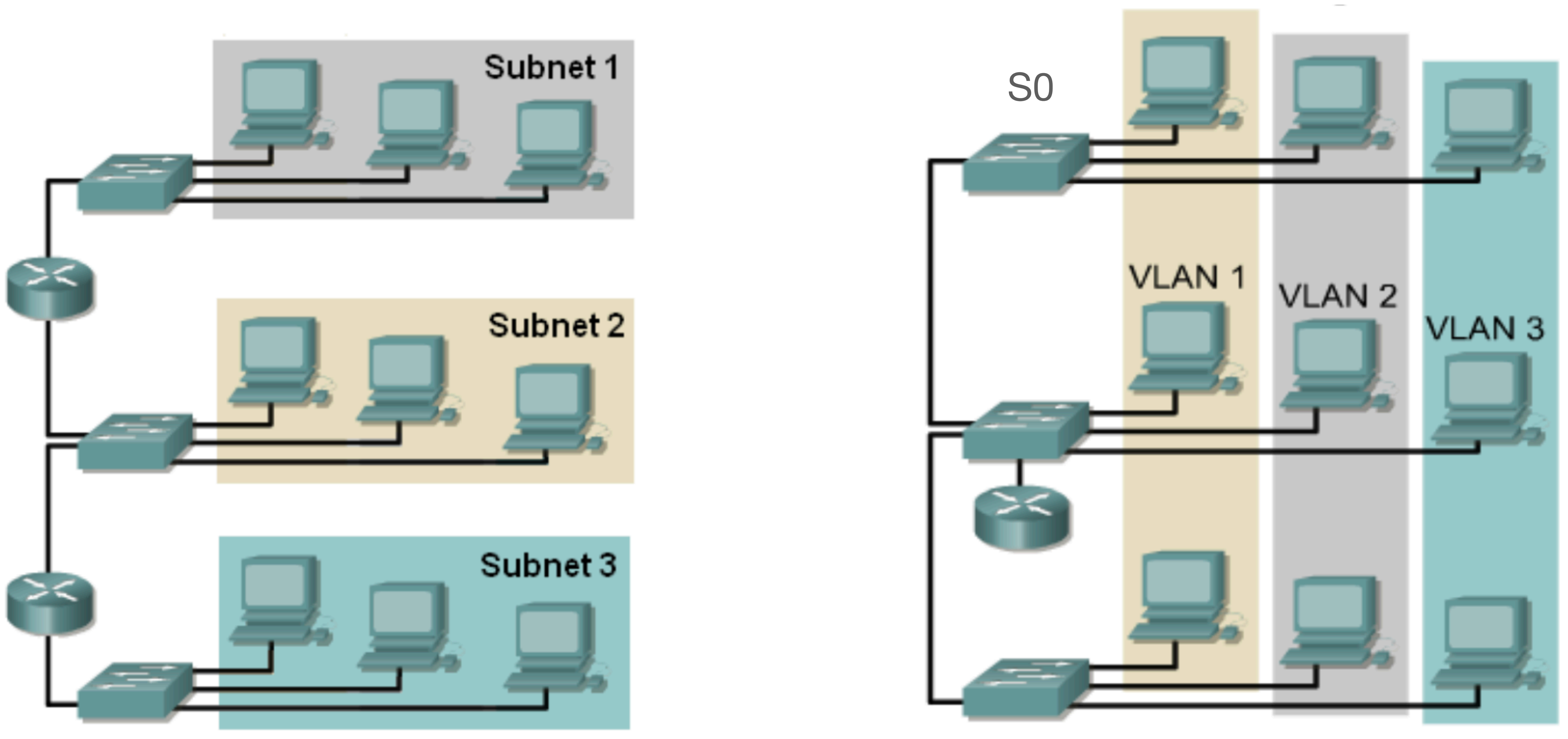
- VLAN spanning means we can connect multiple switches to form a larger network after we use up the ports on existing switches. And the new switch can be set to have the same VLAN.



VLAN SPANNING

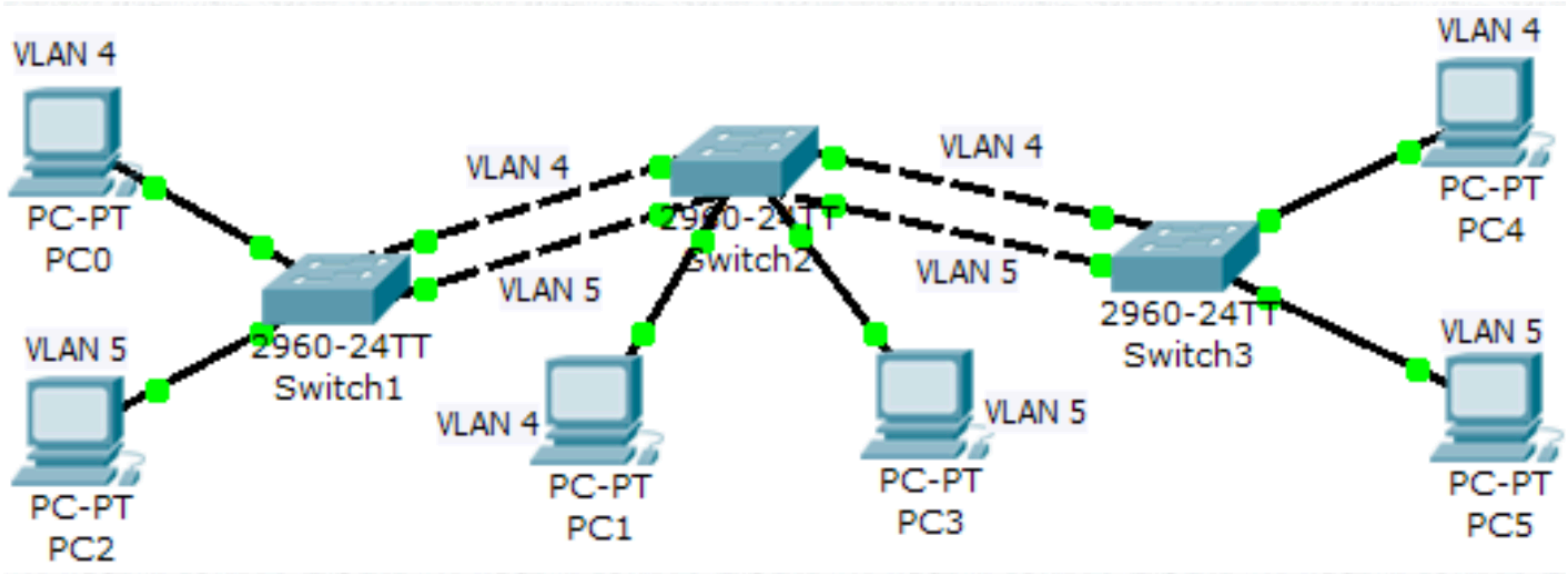
- The idea is that switches are no longer bound to a physical location.
- For example, S0 can has 3 VLAN: VLAN1, VLAN2, VLAN3
- The PC connected to the same switch, like S0, can be from different VLANs.

- Previously, all devices connected to the same switch belongs to the same LAN.
- Now, we must check the VLAN configs on the switch to tell what VLAN the connected device belongs to.



VLAN SPANNING (EXAMPLE)

We have VLAN4 and VLAN5 connected across 3 switches.
Only the PC in the same VLAN can ping each other.

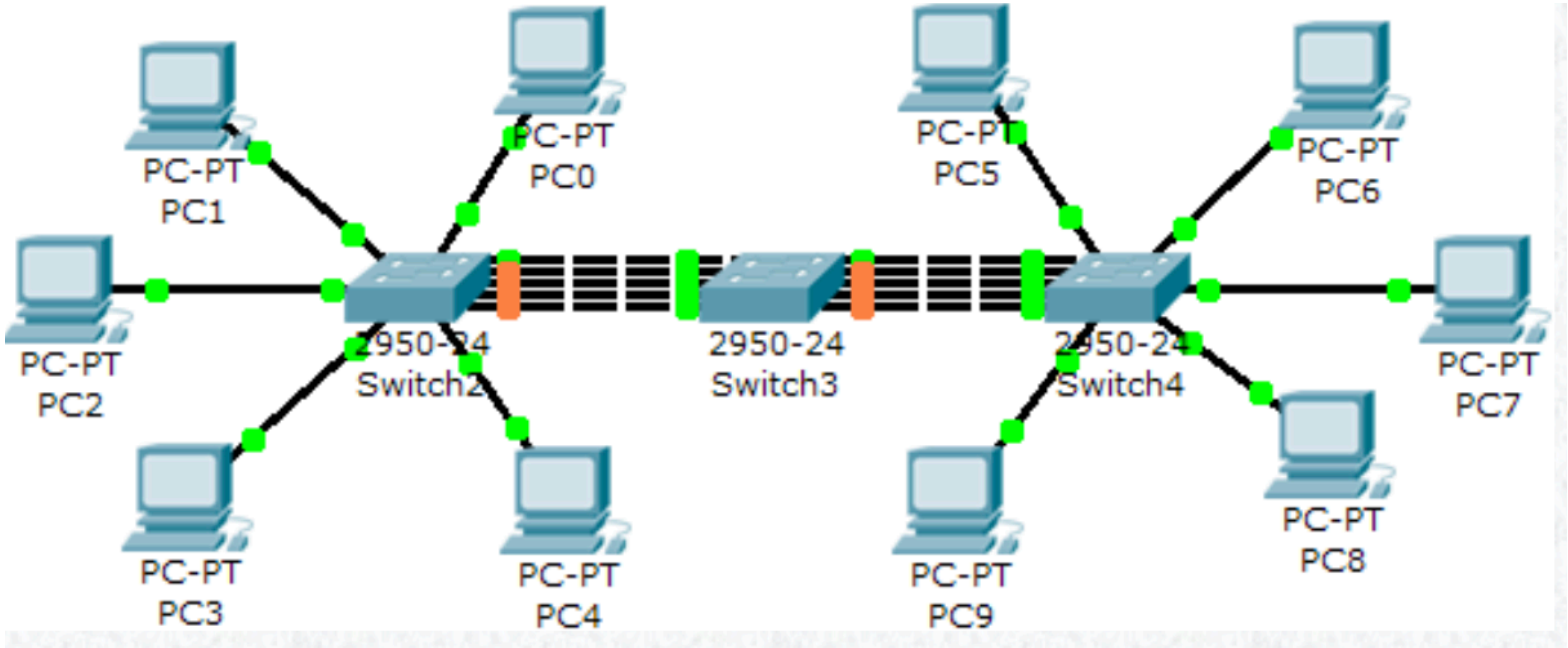


There are two problems:
(a) VLAN4 cannot ping VLAN5
(b) Each VLAN requires 1 cable to connect to the switch

VLAN SPANNING (EXAMPLE)

Imagine if we have:

5 VLAN in one switch

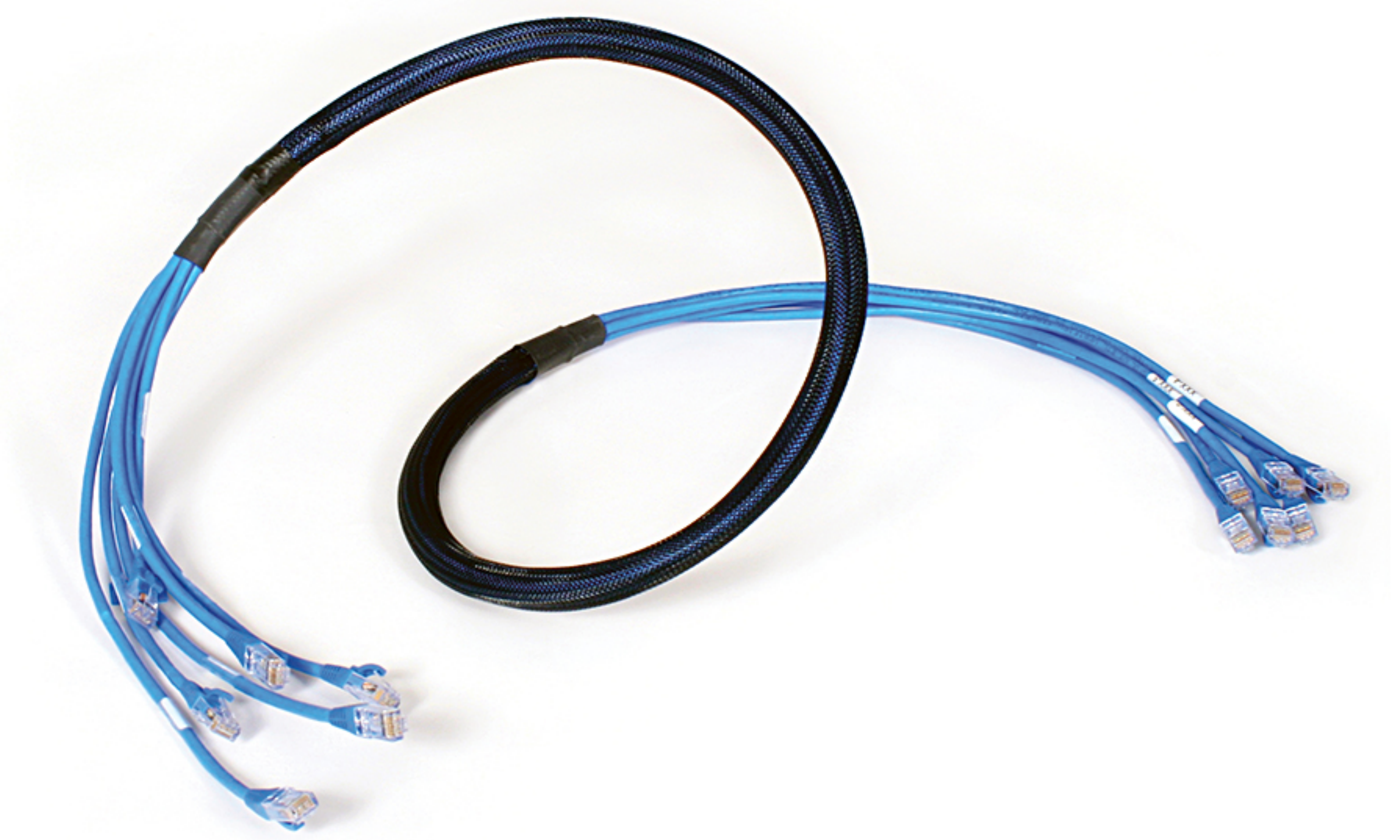


1 line for 1 VLAN, so we have 5 lines on each switch
Pity that the middle switch needs 5x2 = 10 ports used for switch to switch connections. That means we do not have many ports left for PC

TRUNKING

TRUNKING

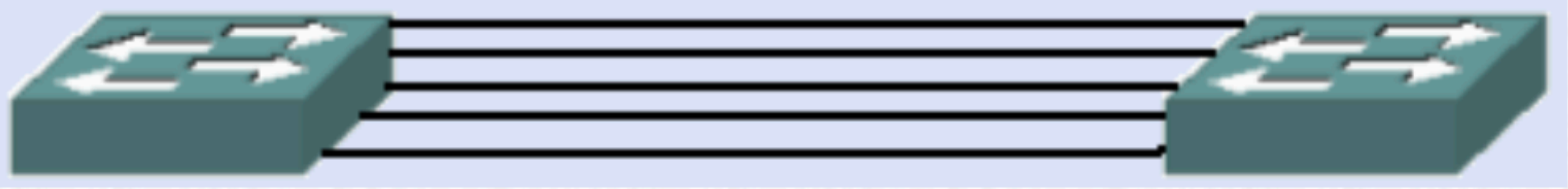
ONE CABLE TO RULE THEM ALL



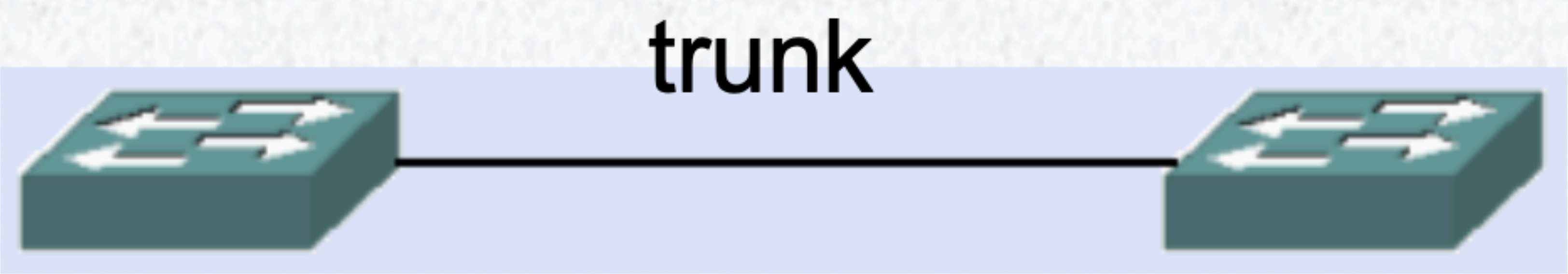
Allow us to use one cable to link multiple VLAN
We need to set TRUNK mode configuration

TRUNKING

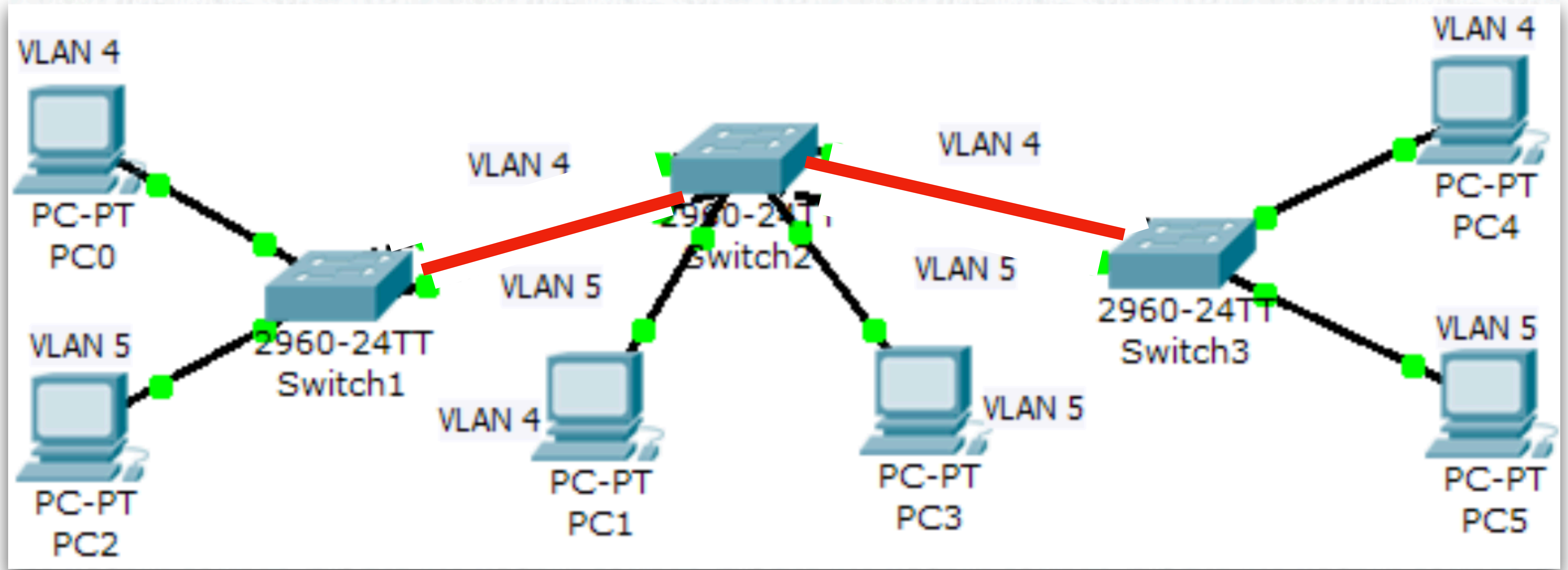
Both switches have the same 5 VLANs.
You need 5 cables to link them up.



However, by using a technique called VLAN trunk, we can combine 5 cables (or many cables) into 1 single cable.



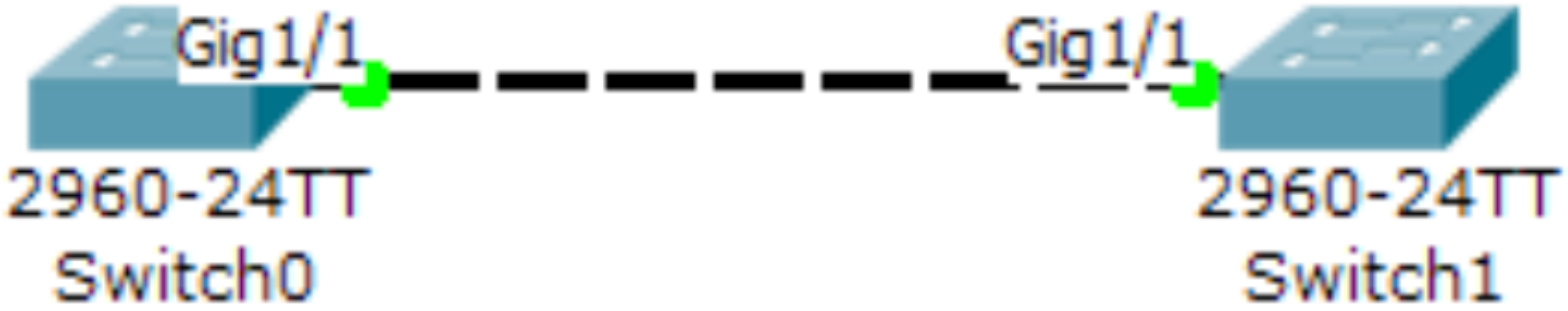
TRUNKING



- Trunk connection allows data from multiple VLANs to travel across a single link
- Reduce the number of cables needed
- The red line can carry traffic from VLAN4 and VLAN5

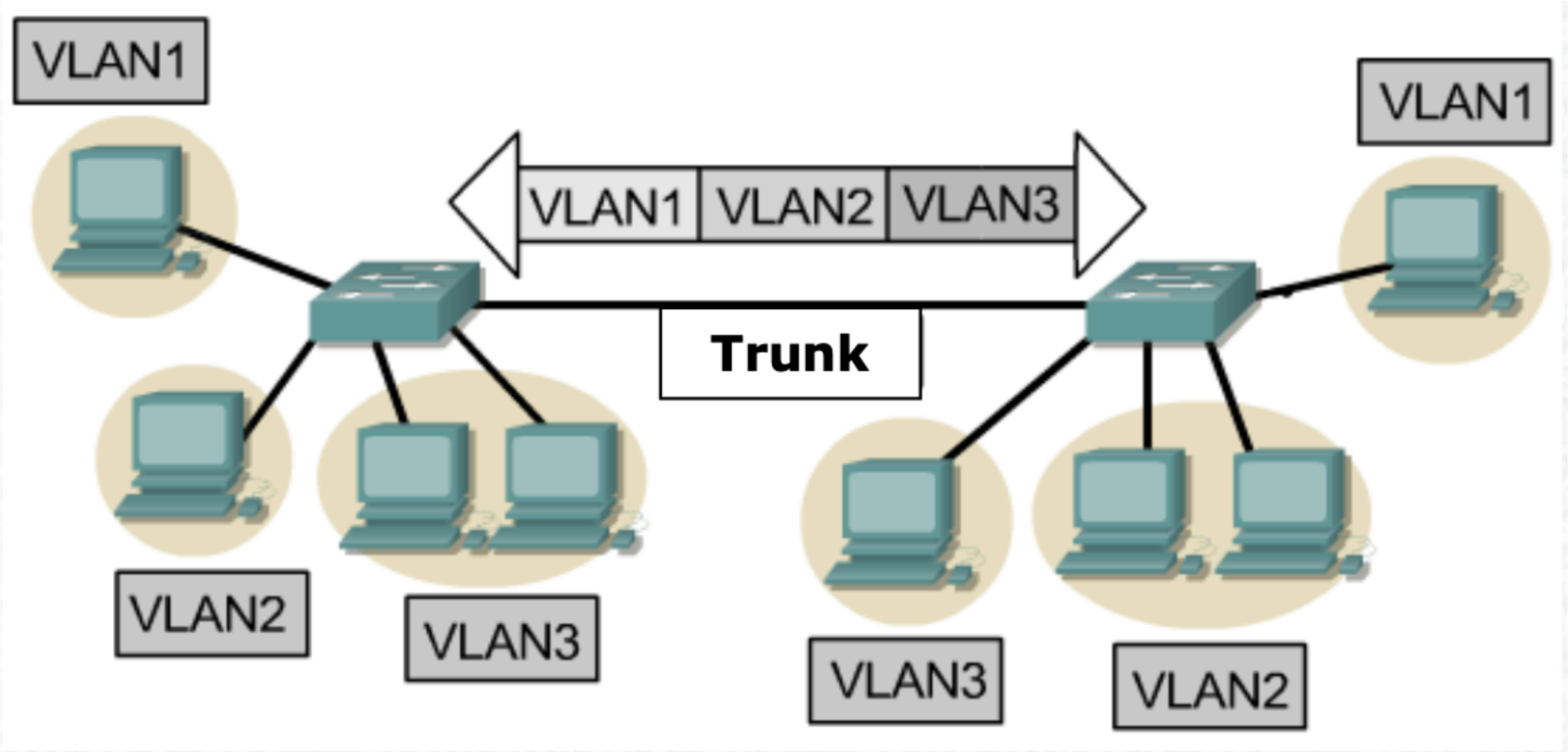
TRUNKING (How to set)

```
Switch(config)#int gig1/1
Switch(config-if)#switchport mode trunk
```




```
Switch(config)#int gig1/1
Switch(config-if)#switchport mode trunk
```

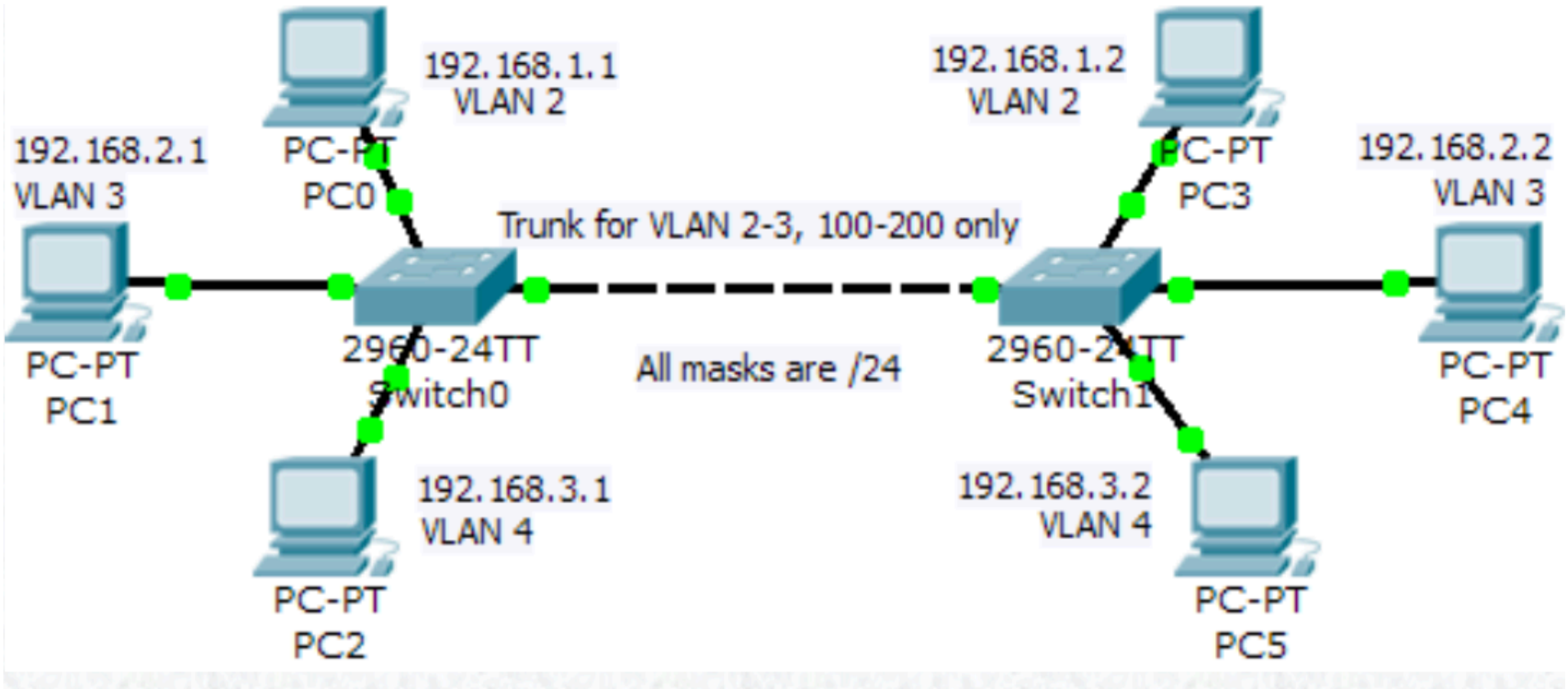
TRUNKING



- The trunk allows traffic from VLAN1, 2, 3 to cross

TRUNKING

- Trunk is like a highway
- We can control which VLAN can cross the trunk cable
- In this example, PC2 cannot ping PC5 because VLAN4 cannot use TRUNK




```
Switch(config-if)#int gig1/1
Switch(config-if)#switchport trunk allowed vlan 2,3,100-200
```

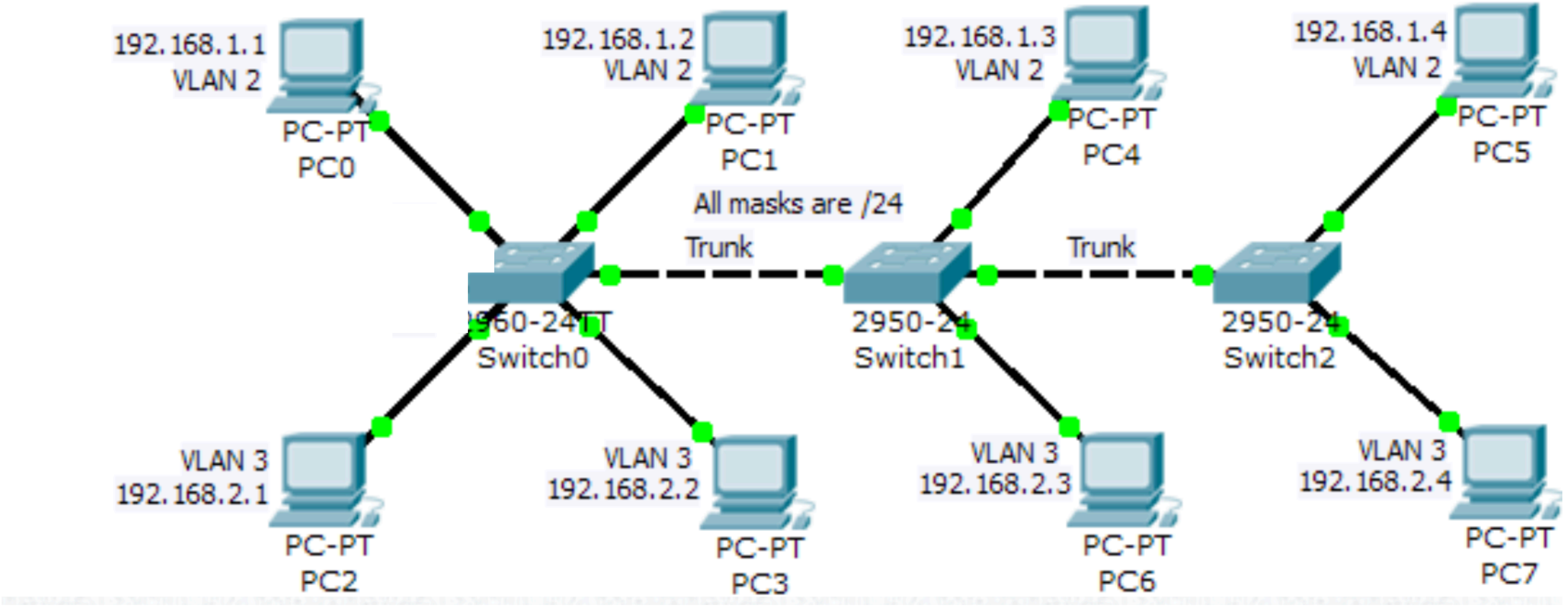
INTERVLAN–ROUTING

INTER-VLAN ROUTING

- So far, we learned how to build VLAN with L3 Switches.
- But, these VLANs cannot talk to other VLANs.
- To allow different VLANs to cross-talk, we need to use a

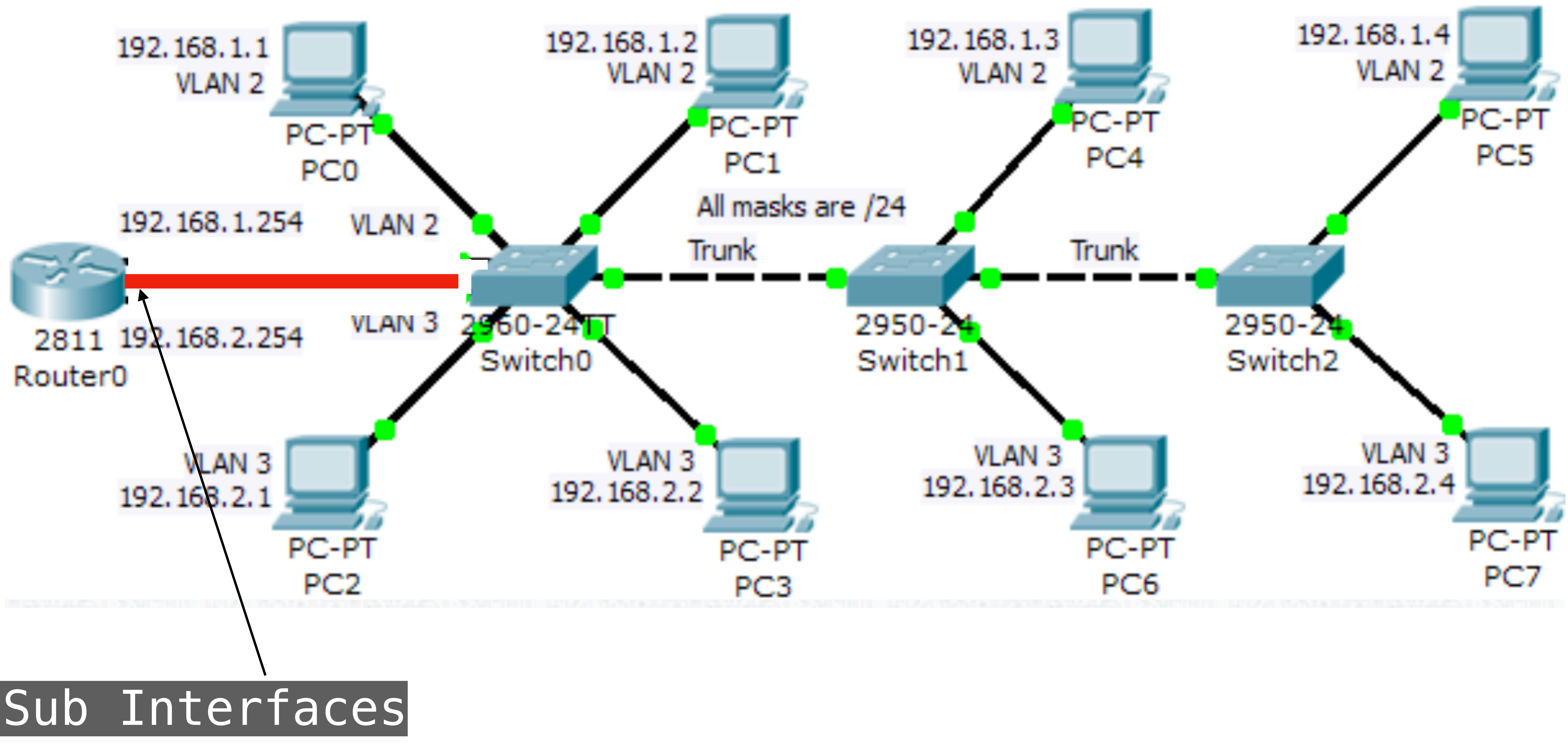
ROUTER

(hi it’s me again)



INTER-VLAN ROUTING (PRO MAX)

- We can apply the same trunk concept here. So that we can use 1 cable to connect the router and the switch.

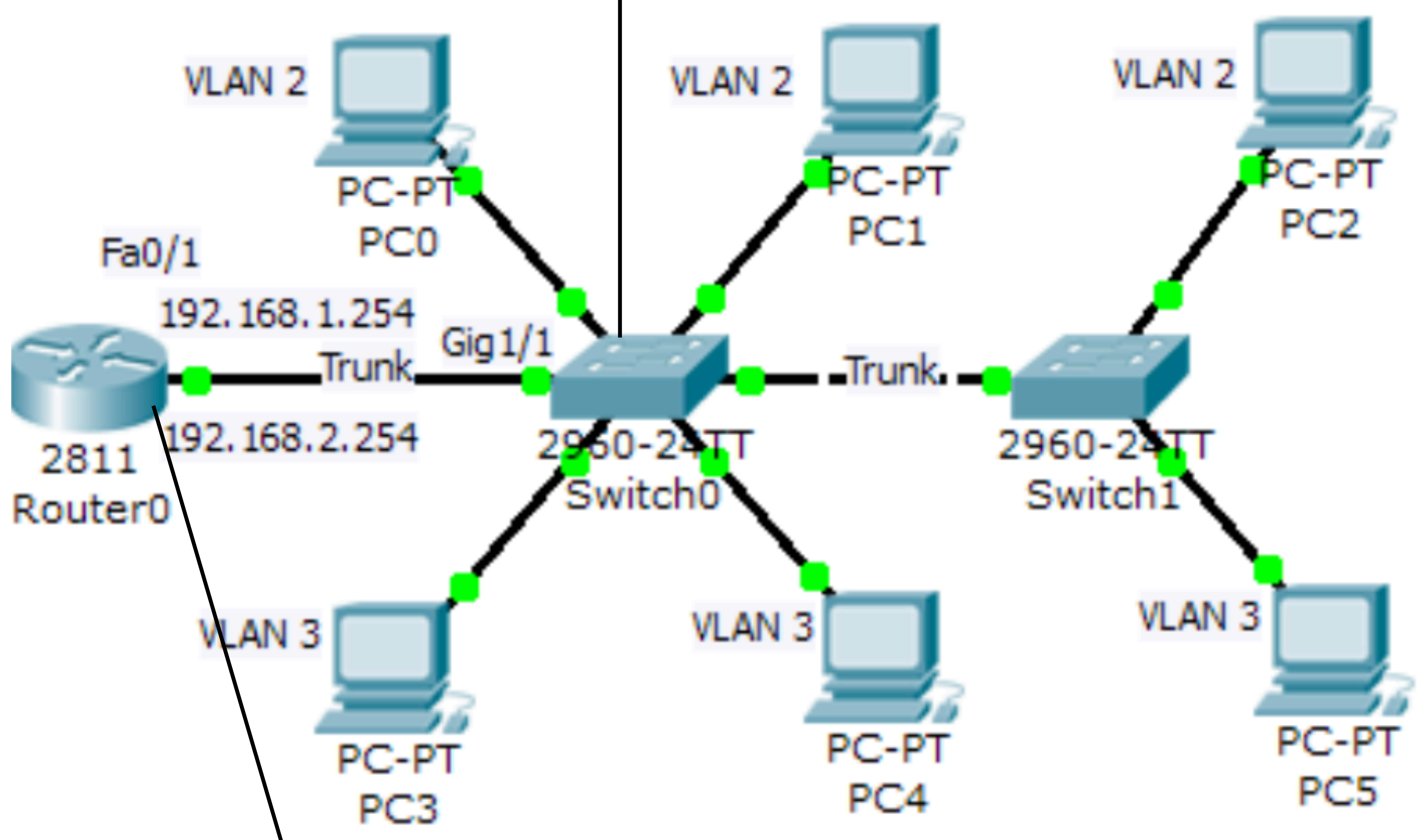


Dot1Q SubInterfaces

- Subinterfaces take the interface name followed by a dot and a number.
  - e.g. `interface f0/0.10`
- It is normal to use the VLAN number. If this ties in with the IP address, even better.
  - e.g. `interface f0/0.2, interface fa0/0.3`
  - ".2" and ".3" are numbers that differentiate the subinterfaces.
- Each of the subinterfaces have an IP address.
- Subinterfaces need to use 802.1Q trunking protocol to differentiate the VLANs in the trunks
  - e.g. encapsulation dot1q 11
  - The above command means only accept traffic of VLAN 11
- The physical interface has no IP address
  - but physical interface need "**no shutdown**"

Dot1Q SubInterfaces (Example)

```
Switch(config)#int gig1/1
Switch(config-if)#switchport mode trunk
```




```
Router(config)#int fa0/1
Router(config-if)#no shutdown
Router(config-if)#int fa0/1.10
Router(config-subif)#encap dot1q 2
Router(config-subif)#ip addr 192.168.1.254 255.255.255.0
Router(config-subif)#exit
Router(config)#int fa0/1.11
Router(config-subif)#encap dot1q 3
Router(config-subif)#ip addr 192.168.2.254 255.255.255.0
```

BENEFITS of VLAN

Benefits of VLAN

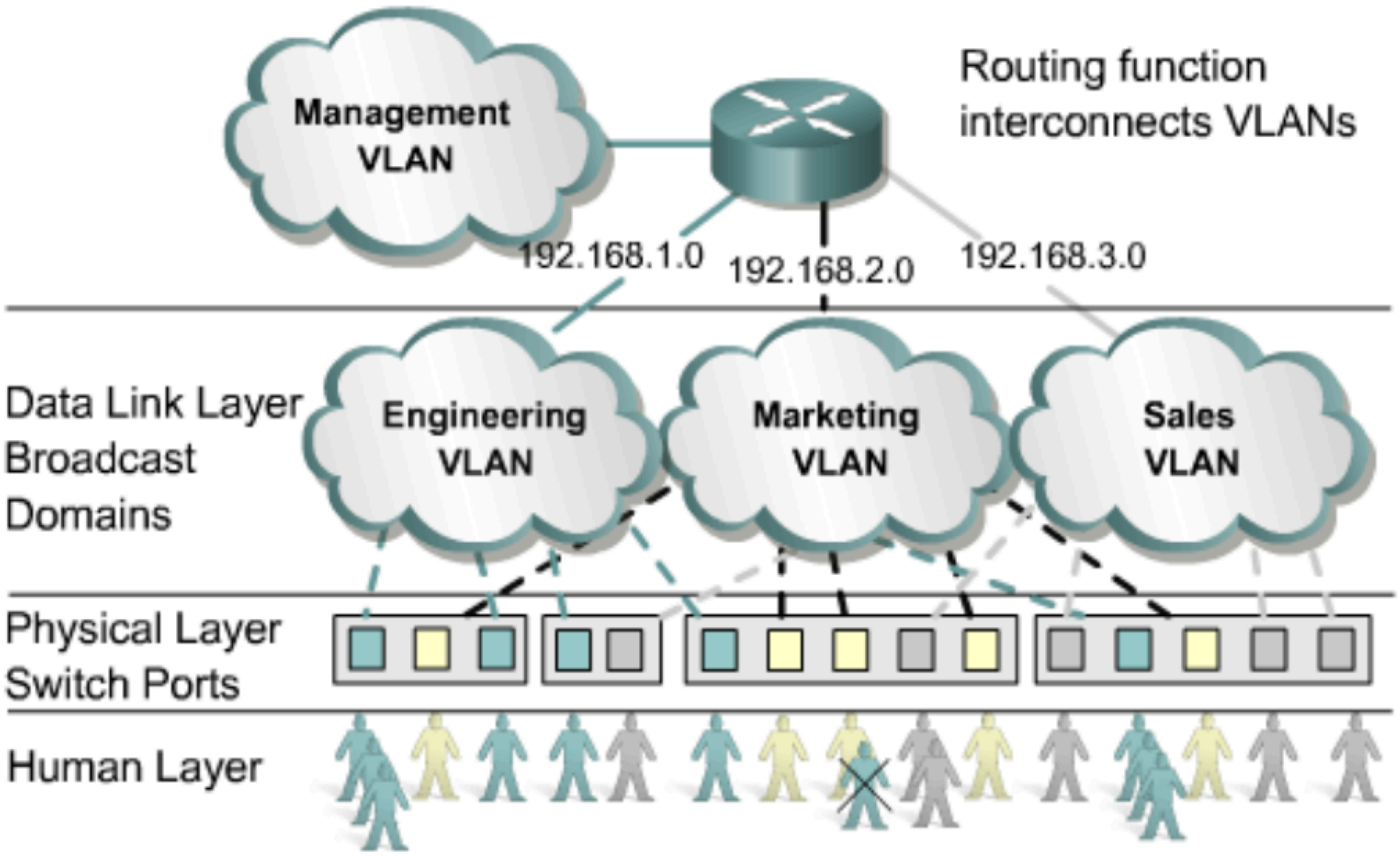
With VLANs, we can improve network security by minimising the impact of broadcast storm attacks.

Broadcast storm is an attack where the hackers spam many packets and flood the network until the network device like router or switches are congested, buffer overflowed and crashed

When we minimise the size of the broadcast domain, the number of affected hosts are reduced.

VLAN SPANNING

With VLAN Spanning, it is easier for users to move around but still join their own VLAN in a different location.

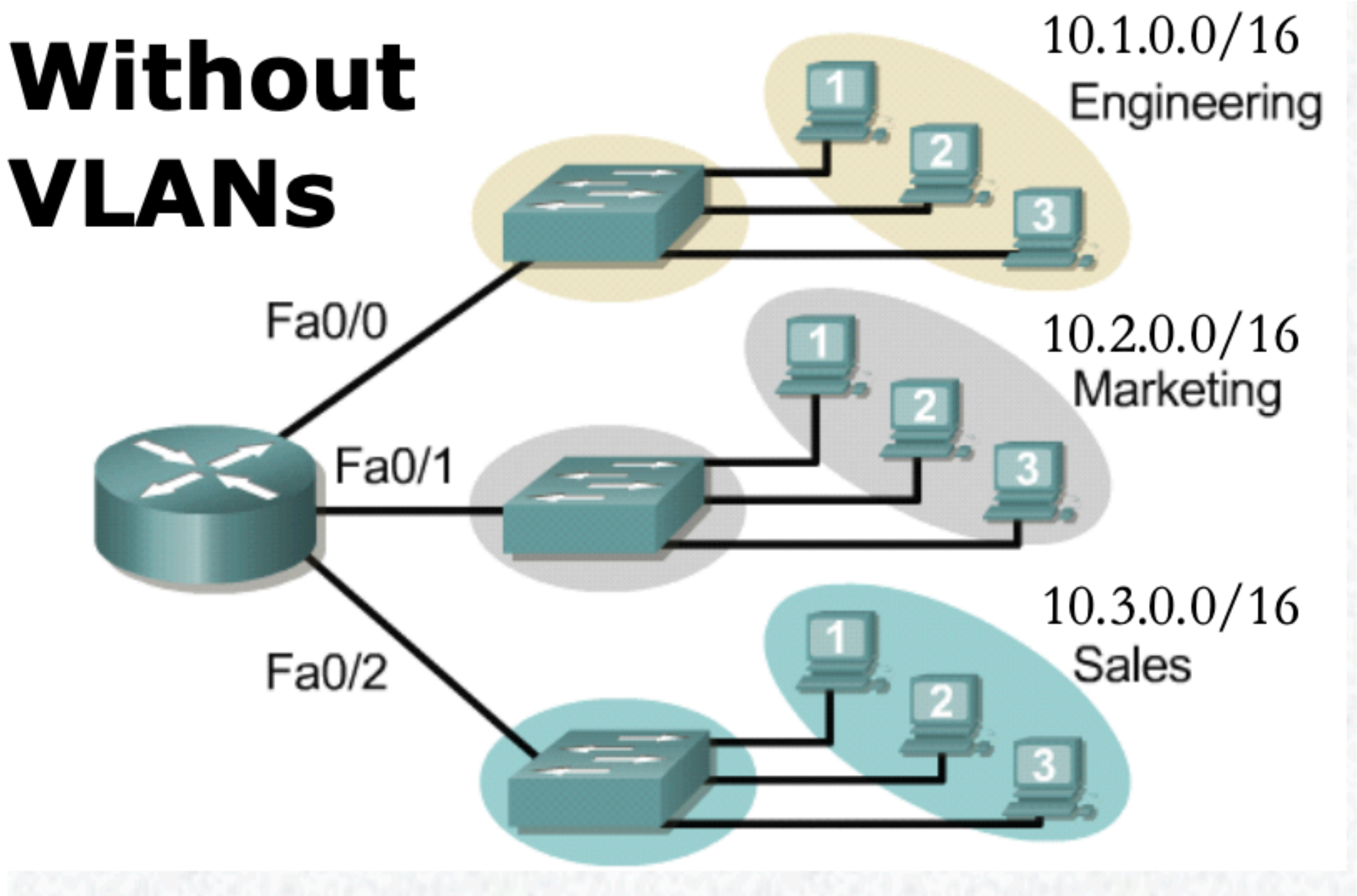


LAN vs VLAN in real world

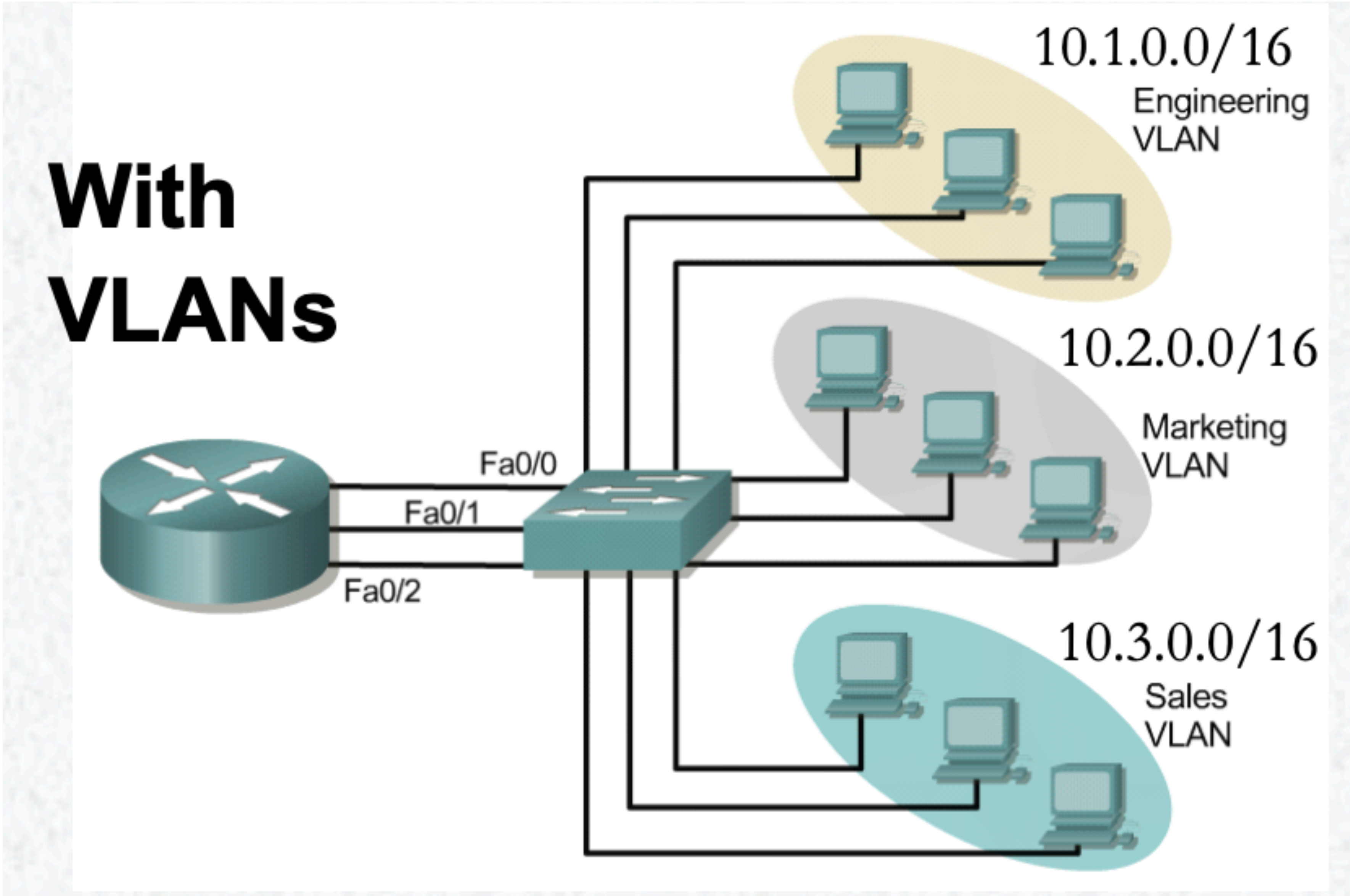
- Now we try to understand two examples from the real world.

- In example 1, each LAN has its own switch. Then, the switch connects to ONE router for global connectivity.

- In example 2, all 3 LANs share one switch. Then, we configure VLAN on this switch to make 3 networks for each department. Finally, we connect this switch to ONE router for global connectivity.



Example1



Example2